

Recognition-Gated Pest-Control Patent & Business Buildout

Roadmap v0.2 - patent-first, conversion-ready provisional strategy

- **Claims:** 135 working exemplary claims across 14 independent families; no ceiling. Claim pruning is deferred to nonprovisional conversion/prosecution strategy.
- **Specification:** draft as a near-nonprovisional disclosure with representative embodiments, drawings, alternatives, failure modes, and explicit support for later claim selection.
- **Principal invention:** audio/video recognition of feral pigs, optional local-population learning and completeness logic, and selective population-control treatment delivery/authorization.
- **Additional branches:** general toxicant, predetermined toxicant, KAPUT®/warfarin, species-selective toxicant, reproductive control, genetic modification, feed, conditioning/trough delivery, system/software, and broader target-animal embodiments.

Patent critical path

- Close inventor-confirmed §112(a) gaps for local-population identity and completeness logic.
- Complete concrete trough/feed/access-control, treatment-controller, safety, logging, and failure-recovery disclosure.
- Resolve species-selective toxicant, reproductive-control, genetic-modification, and other-animal representative support.
- Prepare at least 12 patent figures and integrate figure descriptions with the numbered specification.
- Run claim-to-spec support audit for all independent claims and high-value dependent branches.
- Run element-by-element prior-art charts and formal patent searching; revise without sacrificing supported fallback positions.
- Obtain licensed patent-attorney review and file when disclosure is enabling; claims need not be perfected first.

Conversion strategy

Approximately ten months after the first provisional filing, begin a dedicated conversion review. The objective is to select, prune, consolidate, rearrange, and fee-optimize the claim universe for one or more nonprovisional paths. Any material subject matter conceived after the first provisional should be separately support-mapped and evaluated promptly for a supplemental provisional rather than being saved for the conversion deadline.

Business and funding phase after patent drafting

- Derive product architecture and market segmentation from the protected technical platform and regulatory constraints.
- Evaluate hardware sale, recognition/control software licensing, OEM integration, managed pest-control service, data/monitoring services, and consumables/service revenue where legally appropriate.
- Federal sweep: USDA/APHIS, NRCS, NIFA/SBIR-STTR and other agriculture, invasive-species, AI, conservation, biosecurity, and rural-innovation programs.
- State/local sweep: agriculture, wildlife, conservation, invasive-species, economic-development, university, county/municipal and regional programs; prioritize Mid-Atlantic and Midwest while maintaining nationwide coverage.
- Partnership targets: USDA/APHIS/NWRC, state wildlife/agriculture agencies, universities, farms, landowners, wildlife-control operators, infrastructure sites, and OEM/technology partners.
- International screening: jurisdictions with invasive wild-boar burdens, suitable research partners, and funding/IP terms worth pursuing; separate patent filing strategy from deployment authorization.
- Create NDA-first outreach, pilot protocol, data-rights terms, background/foreground IP terms, publication review, regulatory responsibility, and liability allocation.

Current named technical/regulatory examples

- KAPUT® Feral Hog Bait - EPA Reg. No. 72500-26 - warfarin 0.005% on the accepted EPA label; use only as technically identified patent embodiment and verify current label/status before deployment.

- Sodium nitrite feral-swine toxicant research - useful predetermined-toxicant and trough/pre-baiting technical context; current authorization must be separately verified.
- GnRH/GonaCon™ wildlife reproductive-control research - representative reproductive-control context; do not imply a feral-swine field label without current verification.

Governance

- No public technical disclosure without filing or appropriate NDA.
- Maintain dated conception/inventorship records and the post-filing invention log.
- Tracker and current archive manifest are the project control record; superseded versions are fossils and are excluded from the canonical archive.